

The empirical recurrence for OEIS A230933: recovering a conjecture that is not written down

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Abstract

OEIS A230933 records “Empirical recurrence of order 71”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 328$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 71, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 71 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A230933 is “Number of black-square subarrays of $(n+2) \times (6+2)$ 0..3 arrays $x(i,j)$ with each element diagonally or antidiagonally next to at least one element with value $(x(i,j)+1) \bmod 4$, no adjacent elements equal, and upper left element zero.”. It has offset 1 and begins

42, 438, 3768, 37984, 396950, 4092246, 41991510, 431437274, ...

The entry states:

Empirical recurrence of order 71 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:07:11, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 11697$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 328$ of the 11697, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 328$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 23394$ exact terms of the model returns order 71 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{71} c_i a(n-i),$$

c_1	9	c_{19}	−453524410	c_{37}	29639797784	c_{55}	37296451
c_2	13	c_{20}	−837790459	c_{38}	30932505937	c_{56}	1968949
c_3	36	c_{21}	−719015370	c_{39}	24675633942	c_{57}	−10288430
c_4	49	c_{22}	−132694926	c_{40}	17821487485	c_{58}	−22491492
c_5	−2874	c_{23}	928866705	c_{41}	13562613266	c_{59}	−15839780
c_6	−10563	c_{24}	2864234109	c_{42}	8558622611	c_{60}	−5239165
c_7	−22767	c_{25}	3188534704	c_{43}	3060311673	c_{61}	−1509561
c_8	−69192	c_{26}	3378067991	c_{44}	333223089	c_{62}	1582906
c_9	191852	c_{27}	151425446	c_{45}	−2296212334	c_{63}	1726512
c_{10}	823456	c_{28}	−2241225872	c_{46}	−3755581350	c_{64}	1154851
c_{11}	3720366	c_{29}	−5373723404	c_{47}	−3873668437	c_{65}	279692
c_{12}	10910905	c_{30}	−5480531548	c_{48}	−2559711351	c_{66}	−57532
c_{13}	22173758	c_{31}	−727188313	c_{49}	−1315526527	c_{67}	−106798
c_{14}	46663390	c_{32}	1345260631	c_{50}	−624610795	c_{68}	−38032
c_{15}	55509362	c_{33}	13438624548	c_{51}	−63486304	c_{69}	616
c_{16}	50568210	c_{34}	16331851762	c_{52}	181838049	c_{70}	1616
c_{17}	−21378707	c_{35}	25372994829	c_{53}	175899383	c_{71}	224
c_{18}	−250930649	c_{36}	31452376218	c_{54}	129289482		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 71, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $71 + 4$ terms removed returns order 71 as well, so $\deg q_{\min} = 71 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $71 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 71 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A230933.
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